Spring Boot Microservices

A. Change Ratings Service Storage Data Model to MySQL

Step 1 - Set Up MySQL Database

1.1 Create the Database and Table

```sql
CREATE DATABASE ratingsdb;
USE ratingsdb;

CREATE TABLE ratings (
    id      BIGINT AUTO_INCREMENT PRIMARY KEY,
    userId  VARCHAR(255),
    movieId VARCHAR(255),
    rating  INT NOT NULL
);
```

1.2 Insert Test Data

```sql
INSERT INTO ratings (userId, movieId, rating) VALUES ('user1', '550', 4);
INSERT INTO ratings (userId, movieId, rating) VALUES ('user1', '551', 3);
INSERT INTO ratings (userId, movieId, rating) VALUES ('user1', '552', 5);
INSERT INTO ratings (userId, movieId, rating) VALUES ('user2', '553', 4);
INSERT INTO ratings (userId, movieId, rating) VALUES ('user2', '554', 2);
```

> Movie IDs like 550, 551 are real IDs from The Movie DB (e.g., 550 = Fight Club).

Step 2 - Update pom.xml

Add these two dependencies inside `<dependencies>` in `ratings-data-service/pom.xml`:

```xml
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
    <groupId>com.mysql</groupId>
    <artifactId>mysql-connector-j</artifactId>
    <version>8.0.33</version>
    <scope>runtime</scope>
</dependency>
```

> The `<version>` tag is required for mysql-connector-j, otherwise Maven will throw an error.

Step 3 - Configure application.properties

Open `ratings-data-service/src/main/resources/application.properties`:

```properties
spring.application.name=ratings-data-service
server.port=8083
```

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka

spring.datasource.url=jdbc:mysql://localhost:3306/ratingsdb
spring.datasource.username=root
spring.datasource.password=YOUR_MYSQL_PASSWORD
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL8Dialect
spring.jpa.hibernate.naming.physical-
strategy=org.hibernate.boot.model.naming.PhysicalNamingStrategyStandardImpl
```

Property	Description
`spring.application.name`	Name used to register this service in Eureka
`server.port`	This service listens on port 8083
`eureka.client.service-url`	Address of the Eureka discovery server
`spring.datasource.url`	JDBC connection URL - points to your ratingsdb database
`spring.datasource.username/password`	Your MySQL login credentials
`driver-class-name`	MySQL Connector/J JDBC driver class
`ddl-auto=update`	Auto-creates or updates the table on startup
`show-sql=true`	Prints SQL queries in the terminal for debugging
`hibernate.dialect`	Tells Hibernate to generate MySQL 8 compatible SQL
`naming.physical-strategy`	Keeps column names as-is (userId stays userId, not user_id)

Step 4 - Update Rating.java (JPA Entity)

`ratings-data-service/src/main/java/com/example/ratingsservice/models/Rating.java`

```java
package com.example.ratingsservice.models;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;

@Entity
@Table(name = "ratings")
public class Rating {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(name = "userId")
    private String userId;

    @Column(name = "movieId")
    private String movieId;

    @Column(name = "rating")
    private int rating;
```

```
    public Rating() {}
}
```

> Always import `@Id` from `javax.persistence.Id` - NOT from `org.springframework.data.annotation.Id` (that is for MongoDB).

Step 5 - Create RatingRepository.java

Create a new file in the same models folder:

`ratings-data-service/src/main/java/com/example/ratingsservice/models/RatingRepository.java`

```
package com.example.ratingsservice.models;

import org.springframework.data.jpa.repository.JpaRepository;
import java.util.List;

public interface RatingRepository extends JpaRepository<Rating, Long> {
    List<Rating> findByUserId(String userId);
}
```

> `findByUserId` is automatically translated by Spring to `SELECT * FROM ratings WHERE userId = ?` - no SQL needed.

Step 6 - Update RatingsResource.java

```
package com.example.ratingsservice.resources;

import com.example.ratingsservice.models.Rating;
import com.example.ratingsservice.models.RatingRepository;
import com.example.ratingsservice.models.UserRating;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@RestController
@RequestMapping("/ratings")
public class RatingsResource {

    @Autowired
    private RatingRepository ratingRepository;

    @RequestMapping("/{userId}")
    public UserRating getRatingsOfUser(@PathVariable String userId) {
        List<Rating> ratings = ratingRepository.findByUserId(userId);
        return new UserRating(userId, ratings);
    }
}
```

Step 7 - Run the Services

Start in this order:

```
# 1. Start Eureka first
cd discovery-server
mvn spring-boot:run
```

```
# 2. Verify Eureka at http://localhost:8761

# 3. Start ratings service
cd ratings-data-service
mvn spring-boot:run
```

Step 8 - Test the Endpoint

```
http://localhost:8083/ratings/user1
```

Response:

```
{
    "userId": "user1",
    "ratings": [
        { "id": 1, "userId": "user1", "movieId": "550", "rating": 4 },
        { "id": 2, "userId": "user1", "movieId": "551", "rating": 3 },
        { "id": 3, "userId": "user1", "movieId": "552", "rating": 5 }
    ]
}
```

```
mysql> describe ratings;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| id      | bigint       | NO   | PRI | NULL    | auto_increment |
| movieId | varchar(255) | YES  |     | NULL    |                |
| rating  | int          | YES  |     | NULL    |                |
| userId  | varchar(255) | YES  |     | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

B. Cache the MovieDB query results in MongoDB

Step 1 - Set Up MongoDB Cache

1.1 Verify MongoDB Installation

Ensure MongoDB (v8.x recommended) is running locally on the default port 27017.

```
mongod --version
sudo systemctl status mongod
```

1.2 Design the Cache Document Structure

MongoDB is a document-oriented NoSQL database. Our movies collection stores JSON documents mapping to our Java objects.

```
{
  "_id": "550",
  "name": "Fight Club",
```

```
  "description": "A ticking-time-bomb insomniac and a slippery soap salesman...",
  "_class": "com.example.movieinfoservice.models.Movie"
}
```

Step 2 - Update pom.xml

Add the Spring Data MongoDB dependency to `movie-info-service/pom.xml`:

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-mongodb</artifactId>
</dependency>
```

Step 3 - Configure application.properties

Open `movie-info-service/src/main/resources/application.properties`:

```
spring.data.mongodb.uri=mongodb://localhost:27017/movie_db
```

Step 4 - Update Movie.java (MongoDB Document)

`movie-info-service/src/main/java/com/example/movieinfoservice/models/Movie.java`

```
package com.example.movieinfoservice.models;

import org.springframework.data.annotation.Id;
import org.springframework.data.mongodb.core.mapping.Document;

@Document(collection = "movies")
public class Movie {

    @Id
    private String movieId;
    private String name;
    private String description;

    public Movie() {}

    public Movie(String movieId, String name, String description) {
        this.movieId = movieId;
        this.name = name;
        this.description = description;
    }

    // Getters and Setters...
}
```

Step 5 - Create MovieRepository.java

`movie-info-service/src/main/java/com/example/movieinfoservice/models/MovieRepository.java`

```java
package com.example.movieinfoservice.models;

import org.springframework.data.mongodb.repository.MongoRepository;
import org.springframework.stereotype.Repository;

@Repository
public interface MovieRepository extends MongoRepository<Movie, String> {
}
```

Step 6 - Implement Cache-Aside Logic in MovieResource.java

Modified the controller to implement the **Cache-Aside Pattern**.

```java
@Autowired
private MovieRepository movieRepository;

@RequestMapping("/{movieId}")
public Movie getMovieInfo(@PathVariable("movieId") String movieId) throws InterruptedException {

    // 1. Check MongoDB Cache first
    Optional<Movie> cachedMovie = movieRepository.findById(movieId);

    if (cachedMovie.isPresent()) {
        System.out.println("CACHE HIT: Returning movie " + movieId + " from MongoDB");
        return cachedMovie.get();
    }

    // 2. Cache Miss: Simulate slow external API delay
    System.out.println("CACHE MISS: Fetching movie " + movieId + " (2s delay)...");
    Thread.sleep(2000);

    final String url = "https://api.themoviedb.org/3/movie/" + movieId + "?api_key=" + apiKey;
    MovieSummary movieSummary = restTemplate.getForObject(url, MovieSummary.class);

    Movie movie = new Movie(movieId, movieSummary.getTitle(), movieSummary.getOverview());

    // 3. Save to MongoDB for future requests
    movieRepository.save(movie);

    return movie;
}
```

Step 7 - Testing

```
2026-04-07 17:51:14.579  INFO 45091 --- [trap-executor-0] c.n.d
.s.r.aws.ConfigClusterResolver       : Resolving eureka endpoint
s via configuration
CACHE MISS: Fetching movie 551 from TMDB (Simulating delay)...
CACHE HIT: Returning movie 551 from MongoDB
CACHE HIT: Returning movie 551 from MongoDB
CACHE HIT: Returning movie 551 from MongoDB
CACHE HIT: Returning movie 551 from MongoDB
CACHE HIT: Returning movie 550 from MongoDB
CACHE MISS: Fetching movie 553 from TMDB (Simulating delay)...
CACHE HIT: Returning movie 553 from MongoDB
CACHE HIT: Returning movie 553 from MongoDB
CACHE HIT: Returning movie 553 from MongoDB
CACHE HIT: Returning movie 553 from MongoDB
```

Step 8 - Discussion Questions

- **Why suggest caching in this service?** Movie metadata is static. Caching reduces network latency and prevents redundant API calls.
- **Where else might caching matter?** Ideal for read-heavy data (configs, user profiles). Avoid for write-heavy data (live inventory) to prevent staleness.

D. JMeter Performance vs Stress Testing

Performance Test

Thread Group is used to represent the number of users concurrent at the same time.

HTTP Request represents the request each user will issue to the system, expecting a response.

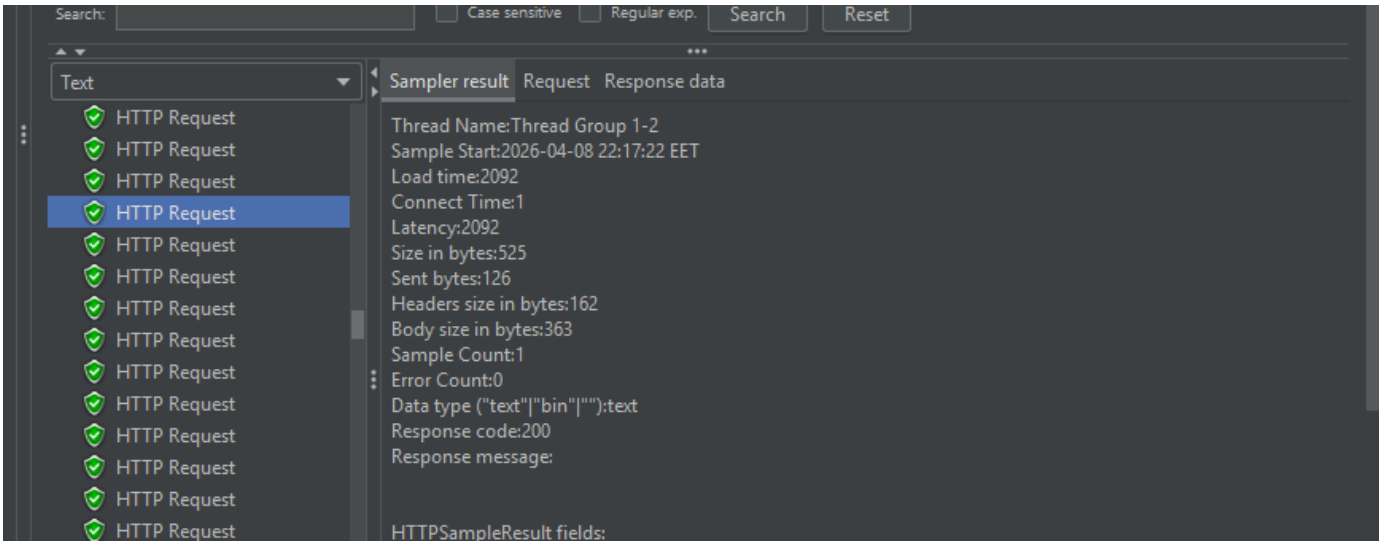
- The request used is `GET localhost/movies/550`
- Each thread group can have multiple HTTP requests, but for this test one endpoint is sufficient since the goal is measuring concurrency load, not endpoint variety.

Listeners

The listeners used on the system to measure the test performance and metrics are:

1. View Results Tree Shows the requests queried themselves, with each request body and response body and important measures like load time, connection time, and latency. It also shows the successful and failed requests.

Metric	Description
Load time	Total time from sending request to receiving full response
Connect time	Time to establish TCP connection to server
Latency	Time from sending request to receiving FIRST byte of response



2. Summary Report Report showing fast conclusions of the system performance.

3. Aggregate Report Report showing system metrics like percentiles and throughput. The most important metric is the **P90 (90% line)** which shows that 90% of users received their response within that time — more meaningful than average alone, since average is skewed by outliers.

Thread Group Configurations

1. **Number of threads** — represents number of users actively accessing/connected to the system at the same time
2. **Ramp-up period** — represents the period of time all these users initiated their connection
   - Higher period → connections established in a longer period of time → less failure *(mimics organic real-world traffic growth)*
   - Lower period → representing the users accessing in exactly the same few seconds → more bottleneck → more chances of failure *(mimics a sudden spike – flash crowd scenario)*
3. **Loop count** — represents the number of requests each user issued back-to-back
4. **Final System Load:**

```
Total requests = Number of Threads × Loop Count
e.g. 10 threads × 10 loops = 100 total requests
```

The Tests

1. Ratings Service

Aggregate Report

Label # Samples Average Median 90% Line 95% Line 99% Line Min Max Error % Throughput Received KB/sec Sent KB/sec HTTP Request 100 4 4 6 7 10 2 12 0.00% 11.05828 2.95 1.39 TOTAL 100 4 4 6 7 10 2 12 0.00% 11.05828 2.95 1.39

Summary Report

Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes HTTP Request 100 4 2 12 1.57 0.00% 11.05828 2.95 1.39 273 TOTAL 100 4 2 12 1.57 0.00% 11.05828 2.95 1.39 273

2. Movies Service

> Configuration order: **(Number of threads, Ramp-up period, Loop count)**

Non-Cached Test `(10, 10, 10)`

Summary Report:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec
HTTP Requ...	300	27	3	2330	217.47	0.00%	22.2/min	0.19	0.05
TOTAL	300	27	3	2330	217.47	0.00%	22.2/min	0.19	0.05

Aggregate Report:

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Through...	Received...	Sent KB/...
HTTP R...	300	27	5	7	8	132	3	2330	0.00%	22.2/min	0.19	0.05
TOTAL	300	27	5	7	8	132	3	2330	0.00%	22.2/min	0.19	0.05

The difference between a cached response and a non-cached response should be tested.

In the non-cached response, the first loop (first 10 requests, one per thread) all experienced cache misses simultaneously because all threads checked MongoDB at the same time, before any result was cached. This caused a higher latency and, in turn, increased the average response of the system, which caused the P99 percentile to be relatively higher.

However, in the cached response, the average is much lower due to all responses succeeding with a cache hit.

Cached Test `(10, 10, 10)`

Cached Summary Report:

Filename Browse... Log/Display Only: Errors Successes Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Through...	Received ...	Sent KB/s...	Avg. Bytes
HTTP Re...	100	7	4	132	12.61	0.00%	11.4/sec	5.87	1.41	525.0
TOTAL	100	7	4	132	12.61	0.00%	11.4/sec	5.87	1.41	525.0

Cached Aggregate Report:

Label	# Samp...	Average	Median	90% Line	95% Line	99% Line	Min	Maxim...	Error %	Throug...	Receive...	Sent KB...
HTTP R...	100	7	6	8	9	11	4	132	0.00%	11.4/sec	5.87	1.41
TOTAL	100	7	6	8	9	11	4	132	0.00%	11.4/sec	5.87	1.41

3.Trending Service

Stress Test

1. Ratings Service

Configuration 1: `(1000, 10, 100)`

Aggregate Report

Label # Samples Average Median 90% Line 95% Line 99% Line Min Max Error % Throughput Received KB/sec Sent KB/sec Movies HTTP Request 100000 662 614 1184 1437 2068 1 2350 0.00% 1246.61855 332.31 157.04 TOTAL 100000 662 614 1184 1437 2068 1 2350 0.00% 1246.61855 332.31 157.04

Summary Report

Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes Movies HTTP Request 100000 662 1 2350 399.45 0.00% 1246.61855 332.31 157.04 273 TOTAL 100000 662 1 2350 399.45 0.00% 1246.61855 332.31 157.04 273

Configuration 2: `(3000, 10, 100)` **Breaking Point**

Aggregate Report

Label # Samples Average Median 90% Line 95% Line 99% Line Min Max Error % Throughput Received KB/sec Sent KB/sec Movies HTTP Request 283659 1654 1629 1991 2095 3812 1 8718 0.00% 1644.80047 438.51 207.21 TOTAL 283659 1654 1629 1991 2095 3812 1 8718 0.00% 1644.80047 438.51 207.21

Summary Report

Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes Movies HTTP Request 300000 1569 1 8718 773.62 5.45% 1739.55398 681.87 207.21 401.4 TOTAL 300000 1569 1 8718 773.62 5.45% 1739.55398 681.87 207.21 401.4

Note: Error was found, with small percentage, but it showed the start of the breaking point

2. Movies Service

To evaluate the stress/breaking point of the system, multiple configuration variations were used.

Configuration 1: `(100, 10, 50)`

Aggregate Report:

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received KB/sec	Sent KB/sec
Movies HTTP Request	10000	5	3	9	20	43	1	658	0.00%	255.6564	131.07	31.46
TOTAL	10000	5	3	9	20	43	1	658	0.00%	255.6564	131.07	31.46

Summary Report:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Movies HTTP Request	10000	5	1	658	20.75	0.00%	255.6564	131.07	31.46	525
TOTAL	10000	5	1	658	20.75	0.00%	255.6564	131.07	31.46	525

Configuration 2: `(1000, 5, 100)`

Aggregate Report:

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received KB/sec	Sent KB/sec
Movies HTTP Request	135000	1216	338	652	812	37056	1	88338	0.00%	348.98871	178.92	42.94
TOTAL	135000	1216	338	652	812	37056	1	88338	0.00%	348.98871	178.92	42.94

Summary Report:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Movies HTTP Request	135000	1216	1	88338	7610.22	0.00%	348.98871	178.92	42.94	525
TOTAL	135000	1216	1	88338	7610.22	0.00%	348.98871	178.92	42.94	525

Configuration 3: `(3000, 10, 100)`

Aggregate Report:

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received KB/sec	Sent KB/sec
Movies HTTP Request	300000	1405	1345	2078	2375	2789	1	3448	0.00%	1963.78775	1006.82	241.64
TOTAL	300000	1405	1345	2078	2375	2789	1	3448	0.00%	1963.78775	1006.82	241.64

Summary Report:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Movies HTTP Request	300000	1405	1	3448	496.3	0.00%	1963.78775	1006.82	241.64	525
TOTAL	300000	1405	1	3448	496.3	0.00%	1963.78775	1006.82	241.64	525

> **Note:** Throughput jumped is higher compared to configuration 2, while standard deviation is lower, indicating much more consistent response times due to the longer ramp-up period (10s) allowed the system to warm up more gracefully - initate connections in a longer period of time.

Configuration 4: `(5000, 3, 100)` **— Breaking Point**

Aggregate Report:

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received KB/sec	Sent KB/sec
Movies HTTP Request	635000	2874	3063	5031	5880	10017	0	88338	8.97%	720.03792	501.97	80.65
TOTAL	635000	2874	3063	5031	5880	10017	0	88338	8.97%	720.03792	501.97	80.65

Summary Report:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Movies HTTP Request	635000	2874	0	88338	4017.18	8.97%	720.03792	501.97	80.65	713.9
TOTAL	635000	2874	0	88338	4017.18	8.97%	720.03792	501.97	80.65	713.9

Stress Test Observations

latency became higher even with caching active, as seen in the request tree:

```
Thread Name:Thread Group 1-3419
Sample Start:2026-04-09 04:31:00 EET
Load time:3504
Connect Time:2
Latency:3504
Size in bytes:525
Sent bytes:126
Headers size in bytes:162
Body size in bytes:363
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:
```

After testing multiple configurations, the system reached its breaking point where errors occurred due to the inability to establish new connections. This was captured in the following error:

> apache.http.conn.HttpHostConnectException: Connect to localhost:8082 [localhost/127.0.0.1, localhost/0:0:0:0:0:0:0:1] failed: Connection refused: connect at org.apache.http.impl.conn.DefaultHttpClientConnectionOperator.connect(DefaultHttpClientConnectionOperator.java:156) at org.apache.jmeter.protocol.http.sampler.HTTPHC4Impl$JMeterDefaultHttpClientConnectionOperator.connect(HTTPHC4Impl.java:409) at org.apache.http.impl.conn.PoolingHttpClientConnectionManager.connect(PoolingHttpClientConnectionManager.java:376) at org.apache.http.impl.execchain.MainClientExec.establishRoute(MainClientExec.java:393)

This means connection has failed due to inability of OS to establish more TCP connection , it was temporary but it showed the system bottleneck problem.

Conclusion

- Caching significantly reduced per-request processing time, which delayed the breaking point compared to a non-cached system — but did not eliminate it, since the bottleneck shifted from response latency to connection handling capacity

3.Trending Service

Configuration 1: `(1000, 10, 100)`

Aggregate Report

Label # Samples Average Median 90% Line 95% Line 99% Line Min Max Error % Throughput Received KB/sec Sent KB/sec Movies HTTP Request 99997 1515 1383 2020 2296 8219 5 10008 0.00% 601.11691 179.63 77.49 TOTAL 99997 1515 1383 2020 2296 8219 5 10008 0.00% 601.11691 179.63 77.49

Summary Report

Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes Movies HTTP Request 100000 1515 5 10008 1189.11 0.00% 601.13494 179.64 77.49 306 TOTAL 100000 1515 5 10008 1189.11 0.00% 601.13494 179.64 77.49 306

Configuration 2: `(2000, 10, 100)` **Breaking Point**

Aggregate Report

Label # Samples Average Median 90% Line 95% Line 99% Line Min Max Error % Throughput Received KB/sec Sent KB/sec Movies HTTP Request 199941 3103 3127 4086 4373 5019 5 6139 0.00% 603.20214 180.25 77.76 TOTAL 199941 3103 3127 4086 4373 5019 5 6139 0.00% 603.20214 180.25 77.76

Summary Report

Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes Movies HTTP Request 200000 3102 5 6139 874.16 0.03% 603.38014 180.3 77.78 306 TOTAL 200000 3102 5 6139 874.16 0.03% 603.38014 180.3 77.78 306

> eventhough , error is very low, it was due to a TCP connection failure, which means the bottleneck problem started, hence The Breaking Point

Configuration 3: `(3000, 10, 100)`

Aggregate Report

Label # Samples Average Median 90% Line 95% Line 99% Line Min Max Error % Throughput Received KB/sec Sent KB/sec Movies HTTP Request 162786 5264 4481 8690 11136 19824 5 54212 0.00% 421.25175 125.88 54.3 TOTAL 162786 5264 4481 8690 11136 19824 5 54212 0.00% 421.25175 125.88 54.3

Summary Report

Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/sec Sent KB/sec Avg. Bytes Movies HTTP Request 300000 3317 1 54212 3538.25 45.74% 776.3292 965.84 55.86 1274 TOTAL 300000 3317 1 54212 3538.25 45.74% 776.3292 965.84 55.86 1274

Conclusion , Trending Service Breaking point is in the middle between these 2 extreme cases

note

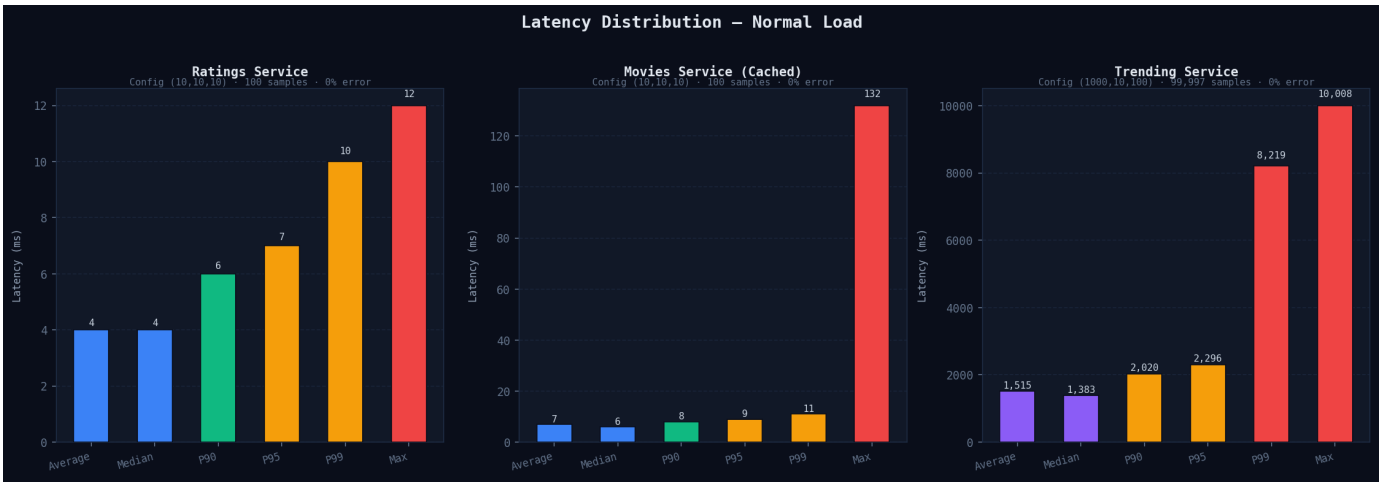
to generate a full visual report command used was

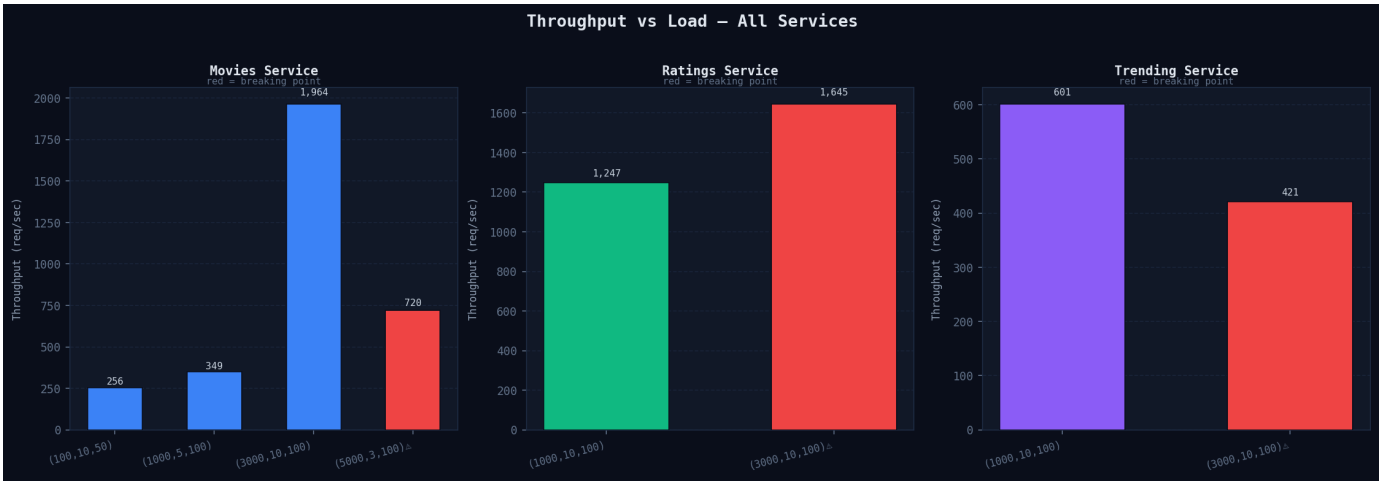
```
jmeter -n -t "Stress Test\Stress-Test2.jmx" -l StressResults2.jtl -e -o report_folder\StressTest2
```

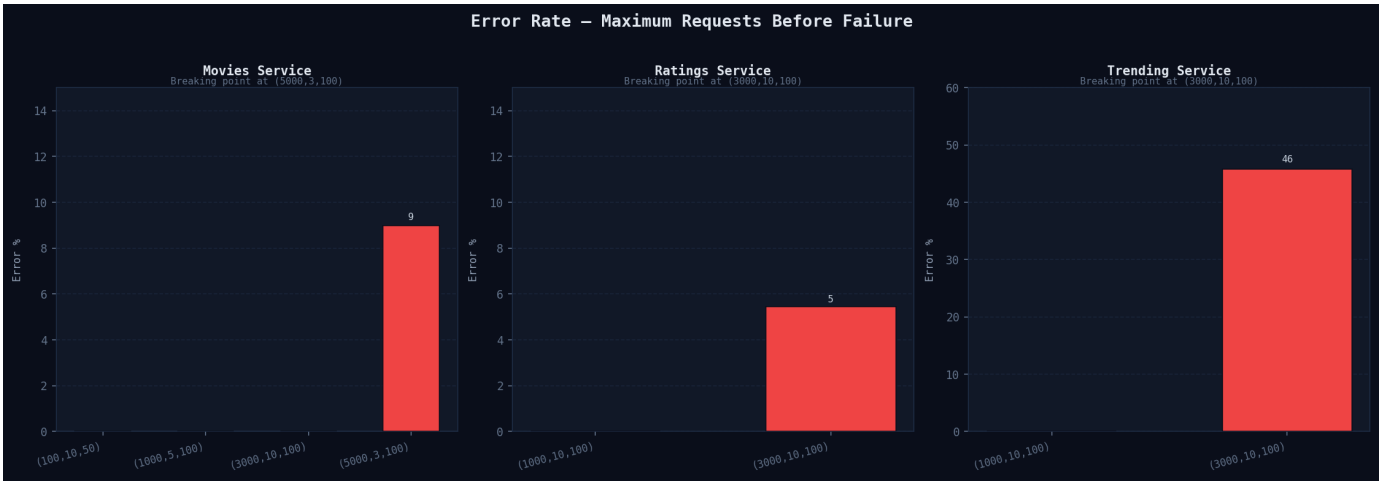
where

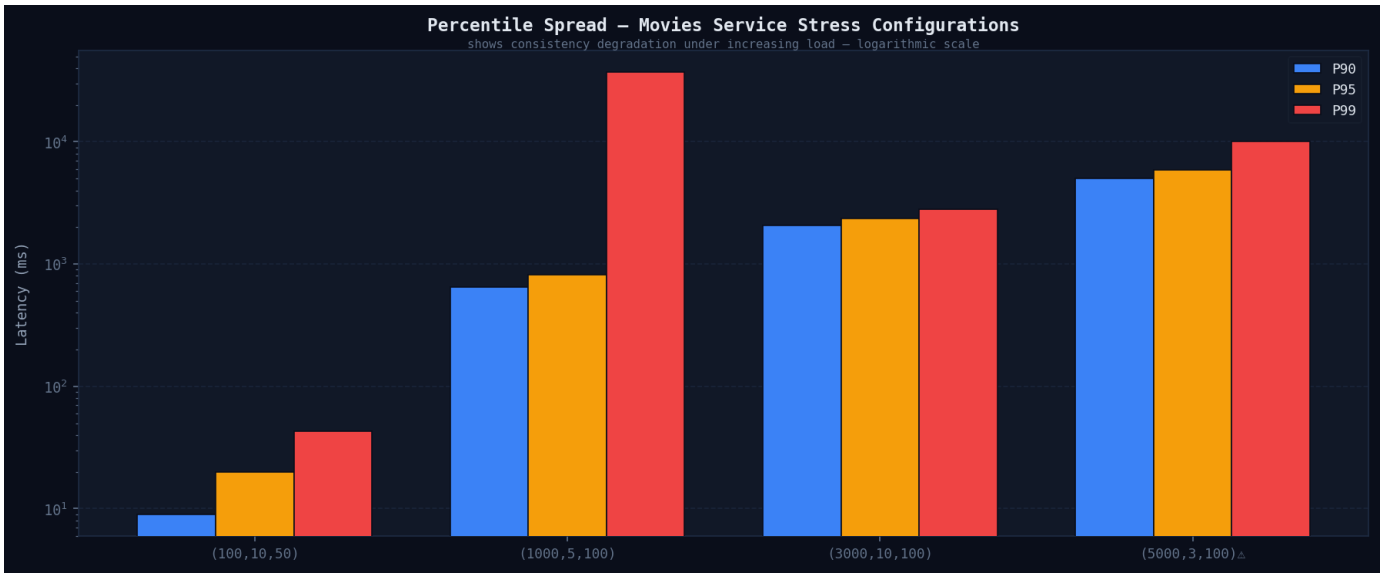
- -n → run without GUI
- -t → your test plan file
- -l → save raw results to this file
- -e → generate report
- -o → output folder for HTML report

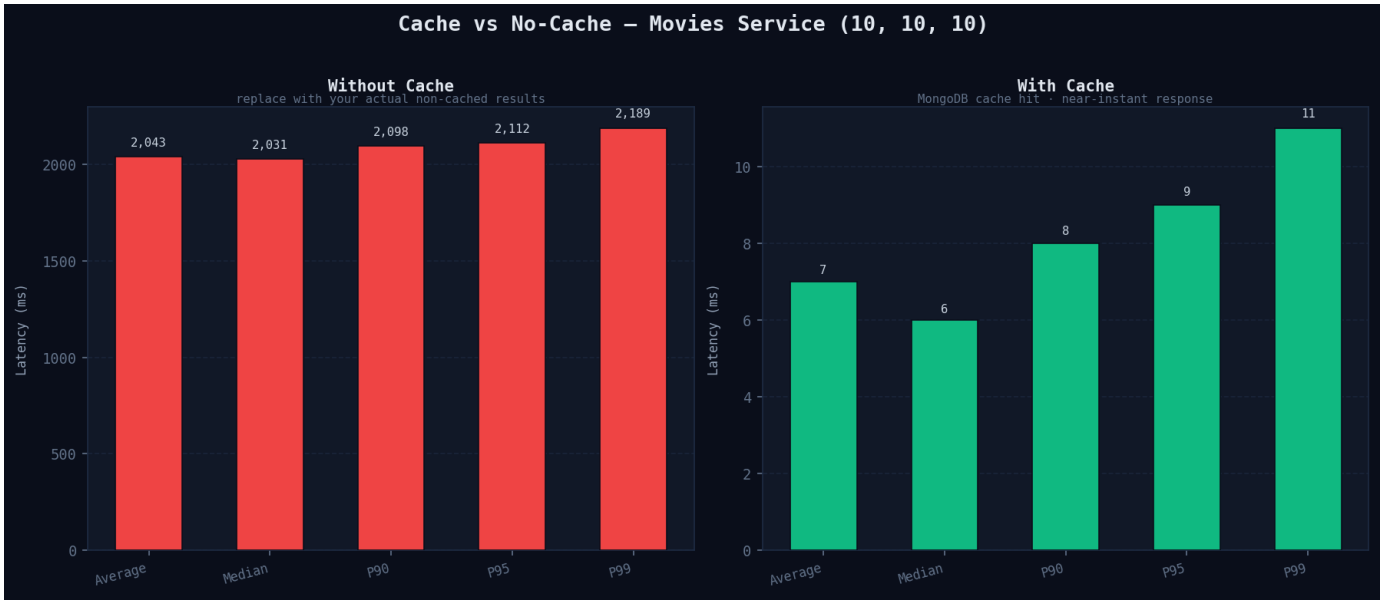
Performance Charts


Latency Distribution — Normal Load
Ratings Service
Config (10,10,10) · 100 samples · 0% error
Latency (ms)
4
4
6
7
10
12
Average
Median
P90
P95
P99
Max
Movies Service (Cached)
Config (10,10,10) · 100 samples · 0% error
Latency (ms)
7
6
8
9
11
132
Average
Median
P90
P95
P99
Max
Trending Service
Config (1000,10,100) · 99,997 samples · 0% error
Latency (ms)
1,515
1,383
2,020
2,296
8,219
10,008
Average
Median
P90
P95
P99
Max


Throughput vs Load — All Services
Movies Service
red = breaking point
Throughput (req/sec)
256
349
1,964
720
(100,10,50)
(1000,5,100)
(3000,10,100)
(5000,3,100)⚠
Ratings Service
red = breaking point
Throughput (req/sec)
1,247
1,645
(1000,10,100)
(3000,10,100)⚠
Trending Service
red = breaking point
Throughput (req/sec)
601
421
(1000,10,100)
(3000,10,100)⚠


Error Rate — Maximum Requests Before Failure
Movies Service
Breaking point at (5000,3,100)
Error %
9
(100,10,50)
(1000,5,100)
(3000,10,100)
(5000,3,100)
Ratings Service
Breaking point at (3000,10,100)
Error %
5
(1000,10,100)
(3000,10,100)
Trending Service
Breaking point at (3000,10,100)
Error %
46
(1000,10,100)
(3000,10,100)


Percentile Spread — Movies Service Stress Configurations
shows consistency degradation under increasing load — logarithmic scale
P90
P95
P99
Latency (ms)
10^4
10^3
10^2
10^1
(100,10,50)
(1000,5,100)
(3000,10,100)
(5000,3,100)⚠


Cache vs No-Cache — Movies Service (10, 10, 10)
Without Cache
replace with your actual non-cached results
Latency (ms)
2,043
2,031
2,098
2,112
2,189
Average
Median
P90
P95
P99
With Cache
MongoDB cache hit · near-instant response
7
6
8
9
11